

Notes: Grossman & Helpman (AER, 1994)

Protection for Sale

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1 Big picture

- **Question.** Why do governments often choose trade policies that deviate from free trade even when free trade is efficient for a small open economy?
- **Core answer.** Trade policy is chosen through a common-agency game between organized industry lobbies and an incumbent government. Lobbies offer contribution schedules contingent on policy. The government trades off campaign contributions against aggregate voter welfare.
- **Key mechanism.** Protection is literally “for sale”: organized sectors bid for tariffs or export subsidies that raise domestic prices for goods in which their specific-factor owners earn rents.
- **Policy result.** Equilibrium protection follows a modified Ramsey rule. A sector receives more protection when it is politically organized, has large domestic output relative to imports, and faces less elastic import demand or export supply.
- **Menu-auction structure.** The trade-policy game has the same structure as the menu-auction/common-agency framework of Bernheim and Whinston. Truthful contribution schedules are focal and generate locally truthful bids around the equilibrium.
- **Contribution result.** The paper characterizes not only the structure of tariffs and subsidies, but also how political surplus is divided between lobbies and the government under truthful Nash equilibria.
- **Instrument-choice result.** Lobbies may prefer inefficient trade policies to more efficient output subsidies because efficient instruments can intensify political competition between organized groups.
- **Economic takeaway.** Trade protection is not an arbitrary reduced-form political distortion. It can be derived as the equilibrium outcome of explicit bids by special interests and an optimizing incumbent government.

2 Incremental contribution of the paper

- **Microfoundation for political-support functions.** Earlier political-economy models often assumed a reduced-form government objective that put exogenous weights on different groups. Grossman and Helpman instead derive the government’s objective from contributions and aggregate voter welfare.
- **Common-agency approach to trade policy.** The paper imports the menu-auction/common-agency toolkit into international trade policy, allowing many lobbies to offer contingent contribution schedules to one policymaker.
- **Explicit protection formula.** The paper delivers a closed-form expression linking equilibrium ad valorem protection to organization status, domestic output-to-import ratios, trade elasticities, and the government’s contribution-welfare tradeoff.
- **Political meaning of cross-industry protection.** The model explains why politically organized import-competing industries with large domestic output and small import volumes receive the strongest protection.
- **Endogenous contribution levels.** The paper goes beyond policy outcomes and studies how much lobbies pay to support them, using truthful Nash equilibria to discipline contribution schedules.
- **Policy-instrument insight.** The paper shows why lobbies need not prefer efficient redistribution. Inefficient instruments can sometimes reduce political rivalry and leave lobbies better off.
- **Empirical agenda.** The modified Ramsey formula became a central benchmark for empirical tests of endogenous protection: protection should vary with import penetration and trade elasticities, conditional on political organization.

3 Model primitives and economic environment

3.1 Economy and goods

- The economy is small and open, so world prices are exogenous.
- There is a numeraire good produced with labor alone and n nonnumeraire goods produced using labor plus sector-specific factors.
- The wage is normalized to one because the numeraire good is produced under constant returns with labor only.
- Domestic prices can differ from world prices because the government chooses import tariffs, export subsidies, import subsidies, or export taxes.

Interpretation: The small-open-economy assumption makes free trade efficient absent politics. Any deviation from world prices is therefore interpreted as politically generated.

3.2 Consumers and indirect utility

Individuals have quasi-linear preferences:

$$U = x_0 + \sum_{i=1}^n u_i(x_i),$$

where x_0 is numeraire consumption and $u_i(\cdot)$ is increasing and strictly concave. Indirect utility is

$$V(p, E) = E + s(p),$$

where $s(p)$ is consumer surplus from the nonnumeraire goods.

Interpretation: Quasi-linearity removes income effects from demand for nonnumeraire goods. This is what makes the trade-policy formula tractable.

3.3 Specific-factor income and trade revenue

Let $\pi_i(p_i)$ be aggregate rents to the specific factor used in sector i , and let $y_i(p_i)$ be domestic output. Per-capita trade tax revenue is

$$r(p) = \frac{1}{N} \sum_i (p_i - p_i^*) [d_i(p_i) - y_i(p_i)].$$

Interpretation: A domestic price above the world price corresponds to import protection or export subsidization, depending on whether the country imports or exports the good.

3.4 Lobby welfare

For organized lobby i , the gross welfare of its members is

$$W_i(p) = \ell_i + \pi_i(p_i) + \alpha_i N [r(p) + s(p)],$$

where ℓ_i is members' labor income and α_i is their population share.

Interpretation: Lobby members care about sector-specific rents, but they also share in consumer surplus and redistributed tariff revenue. This is why lobbies do not always want distortions to be infinitely large.

3.5 Government objective

The government maximizes

$$G(p) = \sum_{i \in L} C_i(p) + aW(p),$$

where $C_i(p)$ is lobby i 's contribution schedule and $W(p)$ is aggregate gross welfare.

Interpretation: The parameter a is the weight on aggregate welfare relative to campaign contributions. A low a means the government is more willing to sacrifice welfare for political contributions.

4 Models and theoretical specifications

4.1 Two-stage political game

1. In stage 1, all organized lobbies simultaneously choose contribution schedules $C_i(p)$.
2. In stage 2, the incumbent government chooses the domestic price vector p to maximize contributions plus weighted welfare.

Interpretation: The model treats campaign contributions as policy-contingent bids. Lobbies do not primarily contribute to change election probabilities; they contribute to buy policy.

4.2 Subgame-perfect equilibrium

The equilibrium consists of contribution schedules and a policy vector such that each lobby's schedule is optimal given the others, and the government chooses the policy that maximizes its own objective.

Interpretation: This is a common-agency problem: several principals (lobbies) try to influence one agent (the government), whose action affects all of them.

4.3 Local truthfulness

A truthful contribution schedule takes the form

$$C_j^T(p, B_j) = \max\{0, W_j(p) - B_j\}.$$

The constant B_j is the welfare level retained by lobby j .

Interpretation: A truthful schedule means that the marginal contribution promised for a policy change equals the lobby's marginal gross willingness to pay for that change.

4.4 Equilibrium policy under truthful schedules

When truthful schedules support the equilibrium, the equilibrium policy maximizes

$$\sum_{j \in L} W_j(p) + aW(p).$$

Interpretation: The government behaves as if it maximizes aggregate welfare plus extra weight on organized interests, because their truthful contribution schedules transfer their policy preferences into monetary bids.

4.5 Modified Ramsey rule

The equilibrium ad valorem trade tax or subsidy in sector i satisfies

$$\frac{t_i^0}{1 + t_i^0} = \frac{I_i - \alpha_L}{a + \alpha_L} \left(\frac{z_i^0}{e_i^0} \right),$$

where $I_i = 1$ if sector i is organized, α_L is the population share represented by lobbies, $z_i^0 = y_i(p_i^0)/m_i(p_i^0)$ is domestic output relative to imports, and e_i^0 is the import-demand or export-supply elasticity.

Interpretation: Organized sectors get positive protection when $I_i = 1$ and α_L is not too large. Protection is stronger when domestic output is large relative to import volume and when demand is inelastic.

4.6 Unorganized sectors

For unorganized sectors, $I_i = 0$, so the formula implies import subsidies or export taxes rather than protection when lobby members are an important population share. If lobby members are negligible, unorganized sectors receive free trade.

Interpretation: Organized lobbies collectively care as consumers about all goods, so they may bid against high prices in sectors where no producer lobby offsets them.

4.7 Truthful Nash equilibrium and contribution levels

The truthful equilibrium pins down policies but not necessarily a unique division of surplus unless further refinement is used. By focusing on truthful Nash equilibria, the paper selects schedules where each lobby pays for policy gains above its retained welfare level B_i .

Interpretation: Contribution levels are determined by political competition among lobbies and the government, not merely by the protected sector's gains.

4.8 Why inefficient trade policies may be preferred

Although output subsidies can transfer income with lower deadweight loss than tariffs, lobbies may prefer trade instruments when efficient instruments intensify opposition from rival lobbies.

Interpretation: Political efficiency differs from economic efficiency. A lobby may prefer an inefficient policy if it lowers the amount it must pay or weakens organized opposition.

5 Identification and theoretical design

5.1 What the theory isolates

- The paper isolates the mechanism by which organized groups convert policy preferences into money bids.

- It keeps the government incumbent and abstracts from electoral competition in order to focus on the fine structure of protection.
- It assumes lobby formation is given, then derives the policy consequences of organization.

Interpretation: The theory is not a theory of which sectors organize; it is a theory of what organized sectors obtain once they can bid for policy.

5.2 Why the menu-auction framework matters

- It disciplines the relationship between contributions and policies.
- It lets lobbies bid over a continuum of policies, not just a discrete alternative.
- It generates the local truthfulness condition that turns lobby willingness to pay into a tractable first-order condition.

Interpretation: Without the menu-auction structure, the government objective would look like an arbitrary political-support function. Here the weights arise from equilibrium bidding.

5.3 Empirical implications

- Protection should be higher in organized sectors.
- Conditional on organization, protection should be higher when the domestic-output-to-import ratio is higher.
- Protection should be lower when import demand or export supply is more elastic.
- The government's willingness to trade welfare for contributions should shift all distortions.

Interpretation: These predictions provide the basis for empirical “protection-for-sale” tests in later political-economy trade research.

5.4 Limitations

- Lobby formation is exogenous.
- The model abstracts from explicit electoral competition.
- The government values contributions in a reduced but interpretable way.
- Corner solutions are mostly set aside.
- The model initially omits downstream users of intermediate goods, though the conclusion discusses this extension.

Interpretation: The model is deliberately stylized. Its strength is the clean mapping from organized interests to equilibrium trade policy.

6 Extracted figures, proposition blocks, and interpretations

6.1 Figure 1: Local Truthfulness + equations (10)–(11): truthful contribution schedules

Read:

- Figure 1 plots a contribution schedule C_i against the domestic price p_i .
- The curve labelled GG is an indifference curve for the government.
- The curve labelled LL is an indifference curve for the lobby.
- A contribution schedule is locally truthful when it is tangent to both relevant indifference curves at the equilibrium.
- Equation (10) defines a truthful schedule: the lobby pays the government for the excess of its gross welfare over a retained payoff B_j .
- Equation (11) states that truthful equilibrium prices maximize the sum of organized lobbies' gross welfares plus weighted aggregate welfare.

Economic message: Local truthfulness is the bridge between the lobby's economic gains and its political bids. It lets the analyst replace arbitrary contribution schedules with marginal willingness to pay.

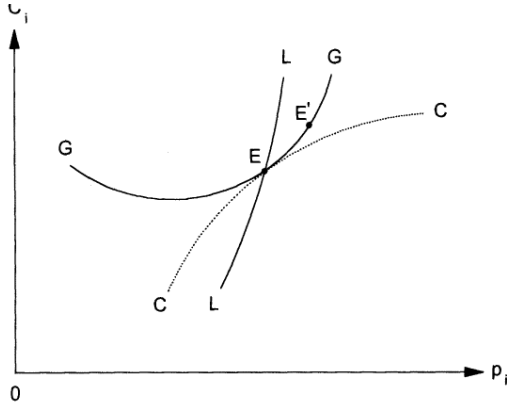


FIGURE 1. LOCAL TRUTHFULNESS

bution schedule so that the marginal change

(a) Figure 1: Local Truthfulness

where, formally, a truthful contribution function takes the form

$$(10) \quad C_j^T(\mathbf{p}, B_j) = \max[0, W_j(\mathbf{p}) - B_j]$$

for some B_j . Notice that truthful schedules are differentiable, except possibly where the contribution becomes nil, because the gross benefit functions are differentiable. Bernheim and Whinston (1986) have shown that players bear essentially no cost from playing truthful strategies, because the set of best responses to *any* strategies played by one's opponents includes a strategy that is truthful. They have also shown that all equilibria supported by truthful strategies, and only these equilibria, are stable to non-binding communication among the players (i.e., they are "coalition-proof"). For these reasons they argue that *truthful Nash equilibria* (those equilibria supported by truthful bid functions) may be focal among the set of Nash equilibria.

Truthful Nash equilibria (TNE) have an interesting property. The equilibrium price vector of any TNE satisfies⁷

(b) Equations (10)–(11): truthful schedules and equilibrium policy

6.2 Proposition 1 (B-W): menu-auction equilibrium + Proposition 2: equilibrium trade-policy formula

Read:

- Proposition 1 gives the Bernheim-Whinston conditions for a subgame-perfect Nash equilibrium of the trade-policy game.
- The key idea is that no lobby can profitably alter its schedule to induce a jointly better policy with the government.

- Proposition 2 delivers the central policy formula: protection depends on organization, lobby population share, government welfare weight, import penetration, and elasticity.
- The formula is a modified Ramsey rule: inelastic sectors are cheaper to distort, so they are more attractive targets for political protection.

Economic message: The first proposition justifies using common-agency logic. The second proposition turns that logic into a testable equation for sectoral protection.

characterized as follows:

PROPOSITION 1 (B-W): $(\{C_i^o\}_{i \in \mathbf{L}}, \mathbf{p}^o)$ is a subgame-perfect Nash equilibrium of the trade-policy game if and only if:

- (a) C_i^o is feasible for all $i \in \mathbf{L}$;
- (b) $\mathbf{p}^o$ maximizes $\sum_{i \in \mathbf{L}} C_i^o(\mathbf{p}) + aW(\mathbf{p})$ on $\mathcal{P}$;
- (c) $\mathbf{p}^o$ maximizes

$$W_j(\mathbf{p}) - C_j^o(\mathbf{p}) + \sum_{i \in \mathbf{L}} C_i^o(\mathbf{p}) + aW(\mathbf{p})$$

on $\mathcal{P}$ for every $j \in \mathbf{L}$;

- (d) for every $j \in \mathbf{L}$ there exists a $\mathbf{p}^j \in \mathcal{P}$ that maximizes $\sum_{i \in \mathbf{L}} C_i^o(\mathbf{p}) + aW(\mathbf{p})$ on $\mathcal{P}$ such that $C_j^o(\mathbf{p}^j) = 0$.

Condition (a) restricts each lobby's contribution schedule to be among those that are

(a) Proposition 1: equilibrium conditions from menu auctions

in terms of the equilibrium ad valorem trade taxes and subsidies, which are defined by $t_i^o \equiv (p_i^o - p_i^*)/p_i^*$.

PROPOSITION 2 (Equilibrium Policies): If the lobbies use contribution schedules that are differentiable around the equilibrium point, and if the equilibrium lies in the interior of $\mathcal{P}$, then the government chooses trade taxes and subsidies that satisfy

$$\frac{t_i^o}{1 + t_i^o} = \frac{I_i - \alpha_{\mathbf{L}}}{a + \alpha_{\mathbf{L}}} \left(\frac{z_i^o}{e_i^o} \right) \quad \text{for } i = 1, 2, \dots, n$$

where $z_i^o = y_i(p_i^o)/m_i(p_i^o)$ is the equilibrium ratio of domestic output to imports (negative for exports) and $e_i^o = -m'_i(p_i^o)p_i^o/m_i(p_i^o)$ is the elasticity of import demand or of export supply (the former defined to be positive, the latter negative).

(b) Proposition 2: equilibrium policies

6.3 Figure 2: Excessive Contributions + Figure 3: Equilibrium Contributions

Read:

- Figure 2 shows why a set of contribution anchors can be excessive: one lobby can raise its retained payoff by shifting the contribution opportunity set without changing the policy actually chosen.
- The shaded region represents the contribution pairs that support a particular policy choice.
- Figure 3 shows an equilibrium configuration in which contribution anchors have been increased until no lobby can improve by raising its retained payoff further.
- The difference between Figure 2 and Figure 3 is about surplus division, not the underlying policy formula.

Economic message: The figures clarify why equilibrium contributions require an additional selection beyond policy optimality. They also show that politics determines both the distortion and who captures the political surplus.

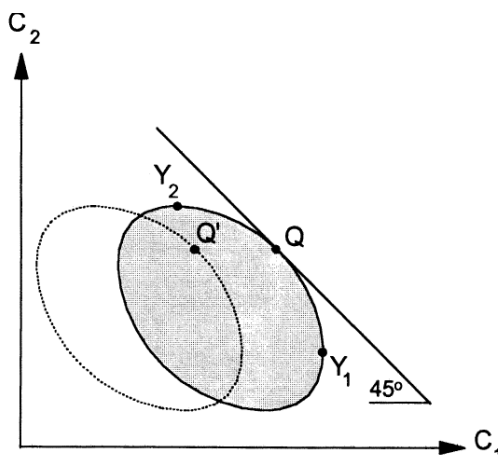


FIGURE 2. EXCESSIVE CONTRIBUTIONS

(a) Figure 2: Excessive Contributions

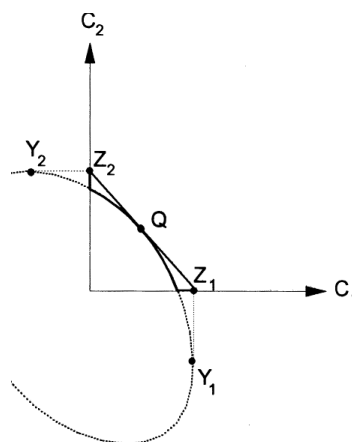


FIGURE 3. EQUILIBRIUM CONTRIBUTIONS

(b) Figure 3: Equilibrium Contributions

6.4 Equations (4)–(6): lobby and government welfare + Section V: why lobbies may prefer trade policies

Read:

- Equations (4)–(6) define lobby welfare, the government's contribution-welfare objective, and aggregate welfare.
- The welfare block shows that lobbies value both producer rents and their shares of consumer surplus and tax revenues.
- The Section V extract explains why efficient instruments such as output subsidies need not be politically preferred by lobbies.
- When efficient instruments increase rivalry among organized groups, lobbies may prefer inefficient trade policies that reduce political competition or shift costs elsewhere.

Economic message: The model distinguishes social efficiency from political equilibrium. Lobbies care about net political payoffs after contributions, not only about deadweight loss.

lects the necessary donations from its members in such a way as to allow all to share in the gains from political coordination.

It will prove convenient in what follows to express the joint welfare of the members of lobby group i as $V_i = W_i - C_i$, where W_i is their gross-of-contributions joint welfare. We note that

$$(4) \quad W_i(\mathbf{p}) = \ell_i + \pi_i(p_i) + \alpha_i N[r(\mathbf{p}) + s(\mathbf{p})]$$

where ℓ_i is the total labor supply (and also the labor income) of owners of the specific input used in industry i and α_i is the fraction of the voting population that owns some of this factor.

The incumbent government cares about the total level of political contributions and about aggregate well-being. The government values contributions, because they can be used to finance campaign spending, and as noted above, they may provide other direct benefits to the officeholders. Social welfare will be of concern to the incumbent government if voters are more likely to reelect a government that has delivered a high standard of living. We choose a linear form for the government's objective function, namely,

$$(5) \quad G = \sum_{i \in L} C_i(\mathbf{p}) + aW(\mathbf{p}) \quad a \geq 0$$

where W represents aggregate, gross-of-contributions welfare.⁵ Aggregate gross and

V. Why Lobbies May Prefer Trade Policies

In deriving the political-economic equilibrium, we have limited the government's choice of policy instruments to trade taxes and subsidies. It may seem that the interest groups would prefer to have the government use more efficient means to transfer income. Our model implies that this is not necessarily the case. In fact, the lobby groups may support institutions that constrain the government to transfer income as inefficiently as possible. Accordingly, a regime that allows only voluntary export restraints (with quota rents transferred to foreigners) may be even more desirable to the lobbies than one that allows for import tariffs. We will discuss now why this is so.¹⁴

Suppose that the government could use output subsidies instead of (or in addition

to import tariffs). A regime that allows output subsidies than in one that does not, because the output subsidies generate less deadweight loss than the trade policies. Moreover, each lobby compensates the government only for the political cost associated with its special treatment (an amount a times the deadweight loss). Therefore, the lobbies capture *all* of the surplus from the use of the more efficient policy instrument.

However, consider now the case where all voters are represented by an organized lobby group. In this situation, as we have seen, the political competition among the groups is quite intense. We know that the equilibrium policy in any TNE maximizes a weighted sum of the utilities of represented and unrepresented voters, and that when all voters are represented in the bidding process the equilibrium policy maximizes aggregate welfare. So the equilibrium entails laissez-faire, just as free trade emerged as the political equilibrium when the government could invoke only trade policies. However, the lobbies must make larger political contributions to induce the laissez-faire outcome in

¹⁴The interested reader can refer to our 1992a working paper for further details.

(a) Equations (4)–(6): lobby and government welfare

(b) Section V: why lobbies may prefer trade policies

7 Short summary

- The paper develops a common-agency model of trade protection.
- Organized industry lobbies offer policy-contingent contributions to an incumbent government.
- The government maximizes total contributions plus a weighted measure of aggregate welfare.
- Truthful contribution schedules make lobbies' marginal bids equal their marginal policy gains.
- Equilibrium protection follows a modified Ramsey rule.
- Organized sectors receive more protection when domestic output is large relative to import volume and when trade elasticities are low.
- The share of voters represented by lobbies moderates protection because lobby members also consume goods and share in tax revenues.
- Contribution levels depend on the division of political surplus between lobbies and the government.
- Lobbies may prefer trade policy to more efficient transfers because efficient transfers can intensify political competition.
- The paper provides a foundational framework for later empirical and theoretical work on endogenous trade policy.

8 Conclusion

8.1 Core conclusion

Grossman and Helpman show that industry protection can be derived from equilibrium political exchange rather than imposed as an exogenous distortion. When lobbies can offer contribution schedules and the government values contributions, trade policy reflects the relative strength of organized interests and the welfare cost of distorting each market.

8.2 Economic intuition

Protection is valuable to a lobby when a higher domestic price raises specific-factor rents. It is costly to the government because it lowers aggregate welfare. It is also costly to other organized interests because their members consume the protected good and pay the associated fiscal costs. Equilibrium protection balances these forces. The resulting tariff is high when the lobby's gain per unit of distortion is large and the welfare loss per unit is small.

8.3 Incremental contribution

The paper's incremental contribution is to transform political-support explanations of trade policy into a microfounded common-agency model with explicit contribution schedules, equilibrium donations, and a testable modified Ramsey formula for protection.

8.4 For applications

The main empirical message is that protection is predicted by organization, import penetration, and elasticities. The main theoretical message is that political power is endogenous to the ability to bid. The main institutional message is that lobbies may favor inefficient policy tools when those tools improve their bargaining position in the political game.

8.5 Limitations and extensions

The framework does not explain why some sectors overcome collective-action problems while others do not. It also abstracts from electoral competition and from organized downstream users in the baseline model. The paper explicitly points to extensions involving intermediate inputs and international policy games. These extensions are important because they move the model closer to modern trade politics, where producers oppose other producers and domestic lobbying shapes international negotiations.

8.6 Broader takeaway

The paper remains foundational because it gives a compact formal answer to a long-standing puzzle: free trade is efficient, but protection persists because political markets allow concentrated beneficiaries to purchase policies whose costs are dispersed across the electorate. The result is not merely that politics matters, but that the structure of protection has a precise economic logic.